Power BI Engineering Labs — Exercise 5

Customer Churn Risk Analysis | Architecture Blueprint

Purpose: document the current Exercise 5 implementation, its data flow, report outputs, and remaining deployment/security work.

1. Architecture overview

Source	Role in solution
SQL Server — AdventureWorksDW2022 (local instance: localhost\\SQLEXPRESS)	Customer and sales data imported into Power BI Desktop.
Exercise5_ML_Scoring_Dataset_MatchedKeys CSV	30 customer scoring rows with CustomerKey, ChurnProbability, and SpendingCluster.
Power BI data model	Customer dimension, internet sales fact, geography dimension, and ML scoring table connected by relationships.
Power BI report	Customer churn risk dashboard with risk-tier breakdowns and revenue-at-risk indicator.

2. Data model and relationships

• DimCustomer[CustomerKey] → FactInternetSales[CustomerKey]: active many-to-one relationship.

• DimCustomer[CustomerKey] ↔ Exercise5_ML_Scoring_Dataset_MatchedKeys[CustomerKey]: active one-to-one relationship; cross-filter direction set to Both.

• DimGeography is present in the model. A regional security rule was attempted on DimGeography using EnglishCountryRegionName = "United States"; this is a static country filter, not a dynamic user-to-country mapping.

3. Churn scoring and report outputs

Churn risk tiers used by the report: High > 0.65; Medium from 0.35 through 0.65; Low < 0.35.

Risk tier	Customer count	Average churn probability
High	10	0.79
Medium	11	0.5027
Low	9	0.2122
Total	**30**	**Approx. 0.51 overall**

Report visuals include customer count, churn-risk distribution, average churn probability by tier, spending-cluster breakdown, customer details, a risk-tier slicer, and a Revenue at Risk card.

Revenue at Risk interpretation

The current measure sums FactInternetSales[SalesAmount] for customers classified as High risk. The displayed value was approximately 17.30K in the report. This is sales associated with high-risk customers, not a validated forecast of future lost revenue; the report should label and explain it accordingly.

4. Security and validation

• HighRiskCustomers role: filter on the ML scoring table where ChurnRiskTier = "High". View-as testing restricted the customer count to 10; stopping role simulation restored the total to 30.

• Regional_Director role: a country filter was added to DimGeography during testing, but its effect did not correctly propagate to the report. Dynamic regional RLS and user-to-country mapping are not complete.

• Current report visuals showed blank values during some View as: Regional_Director tests. This blueprint does not treat that behavior as a successful regional security validation.

5. Publishing and refresh

The report and semantic model were published to Power BI Service in My workspace as AdventureWorks_Exercise5. Scheduled refresh is not confirmed complete: the SQL Server source is local, and Power BI Service indicated that no gateway was available. A working gateway and data-source credentials are still required before scheduled refresh can be validated.

6. Submission checklist

Deliverable	Status / note
AdventureWorks_Exercise5.pbix	Saved by user in Downloads.
Exercise5_DAXStudio_Log.csv	Exported by user to Downloads; verify that the file contains captured query/performance data.
Exercise5_Architecture_Blueprint.pdf	This document.
Regional RLS validation	Incomplete — requires a correct region-to-user mapping and successful View as tests.
Scheduled refresh	Incomplete — configure a suitable gateway and credentials, then test refresh.

Prepared from the project details and screenshots available during the Exercise 5 walkthrough. Confirm outstanding items before submission.